

VC_SYNC

Overview

vc_sync is a synchronization register cell. It models synchronization effects and performs cross-domain checks. The contents of the cell are different for synthesis (implementation) and verification. The synthesis view of **vc_sync** contains two stages of the multibit registers (bit width defined by parameter). The verification view of **vc_sync** contains synchronization effects modeling and synchronization checkers.

Syntax

```
vc_sync #(bit_width, invert) InstName (clk, reset_n, data_in,  
data_out);
```

Parameters	
bit_width	Defines bit width of synchronization register
invert	By default (invert == 0), in case of input data – clock conflict, random value is introduced for one clock cycle to the second synchronization flip-flop. In order to get more pessimistic results, we can replace random function by the input value inversion (invert == 1)
Interface	
clk	Clock input
reset_n	Active-low reset
data_in	Multi-bit input data (before synchronization)
data_out	Multi-bit output data (after synchronization)

Background

The Clock Domain synchronization bugs are critical in the chip design process. It is difficult to find them, since they heavily depend on the test vector patterns, timing etc. These bugs don't belong solely to the verification domain: they are speeded between design, verification and up to synthesis/STA. It is impossible to uncover and fix these bugs with a functional verification only. This task requires a joint work of the number of teams.

VC_SYNC is an innovative verification component that allows to derive the most from the functional verification finding synchronization bugs.

VC_SYNC provides the following checks:

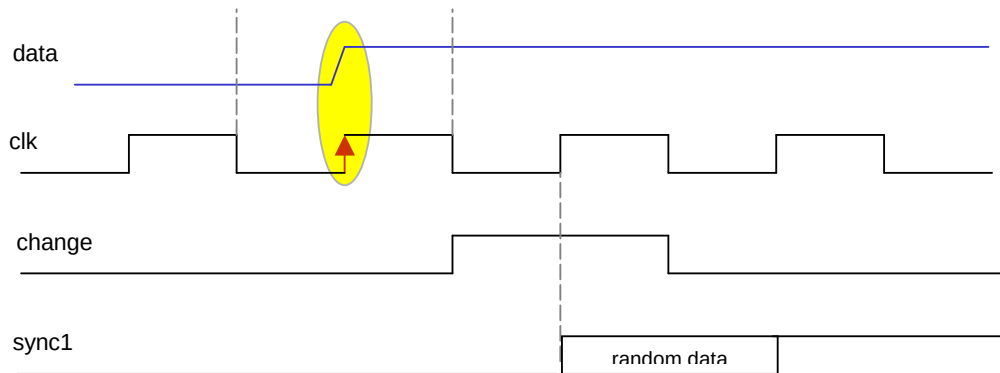
- Single-bit change for the multi-bit signals
- Pulse-width check for the signals crossing fast-to-slow clock domains boundary
- X-value notification

VC_SYNC models the following effects:

- After-synchronization uncertainty of a signal value, on the per-bit basis

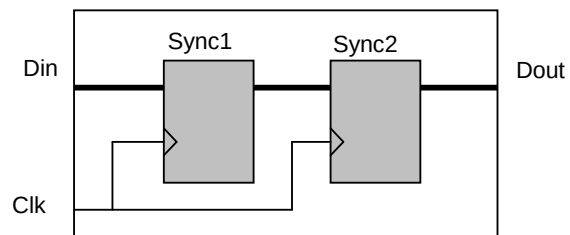
The following figure illustrates this effect:

Figure 1: Metastability effect as simulated by vc_sync component



When VCL_VERIFICATION is defined, a vc_sync component provides all check and simulates all errors as described above. Otherwise, vc_sync contains two stages of the multi-bit parametric registers only (see figure 2). This enables synthesis of the vc_sync component.

Figure 2: Design view of vc_sync component



Verilog Example

```
module test;

reg clk1, clk2;

initial begin
    clk1 = 0;
    clk2 = 0;
end

always #10 clk1 <= ~clk1;
always #12 clk2 <= ~clk2;

reg reg1, reg2, raw_rst_n;
wire reg2_w, rst_n;

always @(posedge clk1)
    reg1 <= $random;

always @(posedge clk2)
    reg2 <= reg2_w;

vc_sync_rst sync_rst (clk2, 1'b0, raw_rst_n, rst_n);
vc_sync sync (clk2, rst_n, reg1, reg2_w);

initial begin
    $dumpvars;
    raw_rst_n = 0;
    #52 raw_rst_n = 1;
    #1000 $finish;
end

endmodule
```